Boris Lin

Senior Software Engineer - Retrieval System, Distributed System, AI Systems
boruilin224@gmail.com | visa sponsorship NOT needed

SKILLS

- Domains: retrieval system (token search/ANN/graph retrieval), distributed system, recommendation system, big data processing, machine learning, RAG
- Frameworks/tools: Apex, Finagle, Airflow, Spark, Kubernetes, AWS, RocksDB, Docker, LangGraph
- Languages: C++, Python, Java, SQL, Scala

WORK EXPERIENCE

Pinterest

Senior Software Engineer • Core Retrieval Infra • 2023 - Present

- Led a group of engineers to develop and deploy a distributed graph-based random walk retrieval system serving terabytes of heterogeneous graph data with tens of milliseconds latency.
- Drove the strategic modernization of a graph-based retrieval system, leading its migration to Kubernetes and establishing a scalable, reliable, and cloud-native serving architecture.
- Developed a scalable, distributed indexing framework (supporting inverted, forward, and ANN indexes) to decouple indexing from serving configurations for independent scaling.
- Built LLM-powered agentic tooling to automate onboarding and debugging workflows, reducing oncall resolution time and improving developer productivity.
- Engineered a core refactor to enable flexible node ID types, supporting graphs with over 4 billion nodes and paving the way for unified embeddings development using Graph Neural Networks.

Senior Software Engineer • Core Serving Infra • 2022 - 2023

- Designed and implemented a unified serving interface to cover all major Pinterest surfaces (Search, Related Pins, Homefeed), ensuring platform consistency and performance.

Software Engineer • Shopping Recommendation Infra • 2020 - 2022

- Built the entire serving architecture to power shopping search result page by integrating a Deep-learned Page Optimization model
- Engineered and deployed comprehensive backend serving infrastructure for Your Shop (Pinterest Shop V1), including candidate retrieval, ranking, blending, caching, and pagination.

Software Engineer • Universal Search • 2018 - 2020

- Rearchitectured Pinterest’s search blending layer into a Universal Search system with features (universal pagination, cache, dynamic timeout) to serve a wide variety of content.
- Developed and shipped Pinterest's first dynamic blending machine learning model (relevance-score based), driving a 70% increase in weighted actions per search — a key milestone for universal search.

EDUCATION

University of Waterloo • Bachelor of Computer Science, Co-op Program • 2013 - 2018